JETHEX
EURO TALENT HUNT EXAM 2026

CLASS : IX Moving to X

euro
JEE – NEET

Time	**: 60 Minutes**
Marks	**: 240**
Medium	**: English**

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on **ORIGINAL Copy** carefully with **blue/black ball point pen only**.

2. The test is of **90 minutes** duration and the Test Booklet contains **60 multiple-choice questions** (four options with a single correct answer) from **Physics, Chemistry, Biology, Mathematics, Mental Ability & Astronomy.**

3. Wherever the symbols/constants are not mentioned, they are to be considered as per their standard meaning/value.

4. Each question carries **4 marks**. For each correct response, the candidate will get **4 marks**.The **maximum marks are 240**.

5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.

6. Rough work is to be done in the space provided for this purpose in the Test Booklet only.

7. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy)** to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.

8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.

9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.

10. Each candidate must show on-demand his/her Admit Card to the Invigilator.

11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.

12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an **Unfair Means case**.

13. Use of Electronic/Manual Calculator is prohibited.

14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination along with Public Examinations (Prevention of unfair means act 2024).

15. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.

16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : …….

Roll Number : in figures ……...

In words ……

Candidate's Signature : …… Invigilator's Signature : ……...

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PART I - PHYSICS

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Q.1 Two resistors of 6 Ω and 3 Ω are connected in parallel across a 12 V battery. If the 3 Ω resistor is disconnected, by what fraction does the total power drawn from the battery decrease?

(A) 1/3 (B) 2/3
(C) ½ (D) 3/4

Q.2 A 1,000 W heater is operated at 200 V. A second heater with twice its resistance is connected to the same supply for the same duration. The heat produced by the second heater is:

(A) twice as much (B) half as much
(C) four times as much (D) one-fourth as much

Q.3 A hydraulic system has piston areas of 5 cm^2 and 200 cm^2. A force of 75 N is applied to the small piston. Ignoring losses, the maximum load balanced by the large piston is:

(A) 3,000 N (B) 150 N
(C) 1,875 N (D) 375 N

Q.4 During a cyclone, the pressure just above a loosely fixed roof drops by 1,500 Pa relative to the indoor pressure. If the roof area is 40 m^2, the net upward force is:

(A) 60 N (B) 600 N
(C) 6,000 N (D) 60,000 N

Q.5 An object is placed 30 cm in front of a concave mirror of focal length 20 cm. The image is formed:

(A) 60 cm in front, real and inverted
(B) 12 cm behind, virtual and erect
(C) 60 cm behind, virtual and erect
(D) 12 cm in front, real and inverted

Q.6 A convex lens forms a real image three times the height of an object on a screen 60 cm from the lens. The object is placed at:

(A) 15 cm (B) 20 cm
(C) 30 cm (D) 180 cm

Q.7 A cyclist covers the first half of a straight road at 12 km/h and the remaining half at 18 km/h. The average speed for the complete journey is:

(A) 15 km/h (B) 14.4 km/h
(C) 16 km/h (D) 13 km/h

Q.8 A body moves from rest with uniform acceleration and covers 24 m in the fourth second alone. Its acceleration is:

(A) 6 m/s^2 (B) 8 m/s^2
(C) 48/7 m/s^2 (D) 12 m/s^2

Q.9 A 4 kg block on a smooth horizontal surface is pulled by two opposing horizontal forces of 30 N and 18 N. Starting from rest, its speed after 5 s is:

(A) 3 m/s (B) 10 m/s
(C) 15 m/s (D) 60 m/s

Q.10 A motor lifts a 40 kg load through 12 m in 8 s. If the motor is 75% efficient and g = 10 m/s^2, its electrical input power is:

(A) 450 W (B) 600 W

(C) 800 W (D) 640 W

PART I - CHEMISTRY

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Q.11 A sealed container contains 10 g of hydrogen and 80 g of oxygen. After a reaction, 90 g of water is claimed to form. Which conclusion is correct?

(A) The claim is possible because reactants are in the required mass ratio
(B) The claim is impossible because hydrogen is in excess
(C) The claim is impossible because oxygen is in excess
(D) The claim violates conservation of mass

Q.12 An atom has atomic number 17 and mass number 37. The number of neutrons in its nucleus and electrons in its neutral state, respectively, are:

(A) 17 and 20 (B) 20 and 17
(C) 37 and 17 (D) 20 and 37

Q.13 Which pair represents isotopes of the same element?

(A) Atoms with 11 protons and 12 neutrons; 11 protons and 13 neutrons
(B) Atoms with 11 protons and 12 neutrons; 12 protons and 11 neutrons
(C) Atoms with 12 electrons and 11 protons; 12 electrons and 12 protons
(D) Atoms with mass number 23 and 24 but different atomic numbers

Q.14 An element X has electronic configuration 2,8,2 and Y has configuration 2,8,7. The most likely formula of their ionic compound is:

(A) XY_2 (B) X_2Y
(C) XY (D) X_2Y_3

Q.15 A 40 g sample of solution contains 6 g solute. If 10 g of water evaporates without loss of solute, the new mass percentage of solute is:

(A) 15% (B) 18%
(C) 20% (D) 25%

Q.16 A mixture contains iron filings, common salt, and sand. Which sequence best yields all three components separately?

(A) Dissolve in water → filter → use magnet on filtrate → evaporate
(B) Use magnet → dissolve remaining mixture in water → filter → evaporate filtrate
(C) Heat strongly → filter → add water → use magnet
(D) Evaporate → distil → use separating funnel

Q.17 A compound always contains 3 g carbon for every 8 g oxygen by mass. How much oxygen must combine with 7.5 g carbon to form the same compound?

(A) 12 g (B) 18 g
(C) 20 g (D) 24 g

Q.18 In the reaction $2Al + 3CuO \rightarrow Al_2O_3 + 3Cu$, which statement correctly follows from the balanced equation?

(A) 2 mol Al reacts with 3 mol CuO to yield 3 mol Cu
(B) 3 mol Al reacts with 2 mol CuO to yield 2 mol Cu
(C) 2 g Al reacts with 3 g CuO
(D) Atoms are created during the reaction

Q.19 Two miscible liquids have boiling points 78 °C and 80 °C. To obtain high-purity fractions, the most suitable method is:

(A) Filtration (B) Simple decantation
(C) Fractional distillation (D) Separating funnel

Q.20 A solute has a solubility of 36 g per 100 g water at a particular temperature. How much additional solute can dissolve in 250 g water already containing 65 g dissolved solute at the same temperature?

(A) 25 g (B) 36 g
(C) 65 g (D) 90 g

PART I - BIOLOGY

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Q.21 A cell is placed in a solution that has a lower water potential than its cytoplasm. Which response is most likely in a living plant cell?

(A) Water enters and the cell becomes turgid
(B) Water exits and the protoplast may pull away from the cell wall
(C) Solutes spontaneously disappear from the vacuole
(D) The rigid cell wall shrinks before the membrane

Q.22 Which observation provides the strongest evidence that an unknown cell is prokaryotic?

(A) It has ribosomes
(B) It possesses a plasma membrane
(C) Its genetic material is not enclosed by a nuclear membrane
(D) It carries out respiration

Q.23 A strip of a young stem can bend without breaking, while a coconut husk is tough and rigid. Which tissues best explain these properties, respectively?

(A) Sclerenchyma and parenchyma
(B) Collenchyma and sclerenchyma
(C) Xylem and collenchyma
(D) Epidermis and meristem

Q.24 A plant has its phloem removed in a complete ring around the stem, while xylem remains intact. Which early result is most likely?

(A) Water transport to leaves immediately stops
(B) Sugars accumulate above the ring
(C) Minerals cannot enter roots
(D) Photosynthesis immediately ceases in all leaves

Q.25 In a pond food chain, algae → zooplankton → small fish → large fish, which is the most likely consequence of a persistent pesticide that biomagnifies?

(A) Highest pesticide concentration in algae
(B) Equal pesticide concentration at every level
(C) Highest pesticide concentration in large fish
(D) Complete absence of pesticide in fish

Q.26 In a closed terrarium kept under continuous bright light, which process directly replenishes the oxygen consumed by animals and decomposers?

(A) Transpiration (B) Photosynthesis
(C) Nitrogen fixation (D) Mineral weathering

Q.27 A person with an infectious respiratory disease is isolated and rooms are kept well ventilated. The main reason ventilation lowers transmission risk is that it:

(A) Kills all microorganisms immediately
(B) Dilutes and removes infectious airborne particles
(C) Increases pathogen mutation rate

(D) Makes all pathogens non-infectious

Q.28 Two groups receive different diets but otherwise similar care. One group develops swollen gums and poor wound healing. Which deficiency most plausibly explains the findings?

(A) Vitamin C (B) Vitamin D
(C) Iodine (D) Iron

Q.29 A microorganism converts atmospheric nitrogen into biologically usable compounds in root nodules of legumes. Its ecological role is primarily:

(A) Decomposer of cellulose (B) Nitrogen fixer
(C) Primary consumer (D) Oxygen producer

Q.30 A muscle cell needs large amounts of ATP for repeated contraction. Relative to a typical fat-storage cell, it is most likely to have:

(A) Fewer ribosomes and no membrane (B) More mitochondria
(C) No nucleus in every case (D) More chloroplasts

PART I - MATHEMATICS

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

31. Simplify : $\frac{2}{\sqrt{7}+\sqrt{5}}+\frac{7}{\sqrt{12}-\sqrt{5}}-\frac{5}{\sqrt{12}-\sqrt{7}}$

(A) 5 (B) 2
(C) 1 (D)0

32. The solution of the two simultaneous equations $2x+y=8$ and $3y=4+4x$ is:

(A) $x=4,\ y=1$ (B) $x=1\ ,\ y=4$
(C) $x=2,\ y=4$ (D) $x=3,\ y=-4$

33. Factorise : $x^2+5\sqrt{3}x+12$

(A) $\left(x+2\sqrt{3}\right)\left(x+3\sqrt{3}\right)$ (B) $\left(x+\sqrt{3}\right)\left(x+4\sqrt{3}\right)$
(C) $\left(x+4\sqrt{3}\right)\left(x-\sqrt{3}\right)$ (D) $\left(x-\sqrt{3}\right)\left(x-4\sqrt{3}\right)$

34. The nearest point from the origin is

(A) $(2,-3)$ (B) $(\ 6,\ 0)$
(C) $(-2,-1)$ (D) (3, 5)

35. The radius of a circle is 6 cm. The perpendicular distance from the centre of the circle to the chord which is 8 cm in length is

(A) $\sqrt{5}$ cm (B) $2\sqrt{5}$ cm
(C) $2\sqrt{7}$ cm (D) $\sqrt{7}$ cm

36. The figure formed by joining the mid-points of adjacent sides of a rhombus is

(A) a square (B) rectangle
(C) trapezium (D) none of these

37. In the sides of a triangle are doubled, then its area :

(A) Remains the same (B) Becomes doubled
(C) Becomes three times (D) Becomes four times

38. In a simultaneous toss of two coins, find the probability of getting two tails.

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{3}{4}$ (D) $\frac{1}{3}$

39. Three cubes with sides in the ratio 3: 4: 5 are melted to form a single cube whose diagonal is 12 $\sqrt{3}$ cm. The sides of the cube are

(A) 6 cm, 8 cm, 10 cm (B) 3 cm, 4 cm, 5 cm

(C) 9 cm, 12 cm, 15 cm (D) 12 cm, 16 cm, 20 cm

40. $\sqrt{\frac{\left(x^2+3x+2\right)\left(x^2+5x+6\right)}{x^2\left(x^2+4x+3\right)}}$ is equal to :

(A) $x\left(x+1\right)$ (B) $\frac{x+2}{x}$

(C) $\frac{x}{x+2}$ (D) $x\left(x+2\right)$

PART II – MENTAL ABILITY

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Q.41 In a code, MANGO is written as OCPIQ, where each letter is shifted two places forward. What is the code for LEMON?

(A) NGOQP (B) NGOQR

(C) MFNPO (D) NGOPQ

Q.42 Complete the series: 3, 8, 18, 38, 78, ?

(A) 148 (B) 156

(C) 158 (D) 160

Q.43 If A > B = C < D and E < C, which statement must be true?

(A) D > E (B) A > D

(C) B > A (D) E = B

Q.44 At 3:40, what is the smaller angle between the hour and minute hands of an analogue clock?

(A) 120° (B) 130°

(C) 140° (D) 150°

Q.45 A cube is painted on all six faces and cut into 64 equal small cubes. How many small cubes have exactly two painted faces?

(A) 16 (B) 24

(C) 32 (D) 48

Q.46 A walks 5 km north, then 12 km east, then 5 km south. How far and in which direction is A from the starting point?

(A) 12 km east (B) 12 km west

(C) 13 km north-east (D) 22 km east

Q.47 In a row of 35 students, Reva is 12th from the left and Zoya is 9th from the right. How many students are strictly between them?

(A) 12 (B) 13

(C) 14 (D) 15

Q.48 A box contains 5 red, 4 blue and 3 green identical balls. Two balls are drawn without replacement. What is the probability both are blue?

(A) 1/11 (B) 1/9

(C) 2/11 (D) 1/12

Q.49 Statement: All comets are celestial bodies. Some celestial bodies are visible without telescopes. Which conclusion necessarily follows?

(A) All comets are visible without telescopes
(B) Some comets are visible without telescopes
(C) All comets are celestial bodies
(D) No comets are visible without telescopes

Q.50 Complete the analogy: 7 : 50 :: 11 : ?

(A) 120 (B) 122
(C) 124 (D) 132

PART II – ASTRONOMY

This section contains **10 Multiple Choice Questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Q.51 A planet is twice as far from the Sun as Earth. Assuming approximately circular orbits, its orbital period is closest to:

(A) 1.4 years (B) 2 years
(C) 2.8 years (D) 4 years

Q.52 A student sees a first-quarter Moon tonight. Approximately how long later should the next full Moon occur?

(A) 2 days (B) 7 days
(C) 14 days (D) 29 days

Q.53 An observer in India sees a star rise about four minutes earlier each successive night. This happens mainly because:

(A) Earth rotates more slowly each day
(B) Earth's revolution changes the direction to the Sun relative to distant stars
(C) The star rapidly moves closer to Earth
(D) The Moon delays sunrise

Q.54 Why does a planet in an elliptical orbit move fastest when it is nearest the Sun?

(A) Its mass increases near the Sun
(B) Equal areas are swept out in equal times, requiring higher speed when nearer
(C) Solar gravity disappears at the farthest point
(D) Its axis tilts more strongly

Q.55 Which observation provides the most direct evidence that Earth is approximately spherical?

(A) Seasons occur on Earth
(B) Different stars are visible from different latitudes
(C) The Sun rises in the east
(D) Wind blows across oceans

Q.56 A telescope's objective diameter is doubled while all else remains comparable. Approximately how does its light-gathering power change?

(A) Doubles (B) Triples
(C) Quadruples (D) Remains constant

Q.57 An astronaut on the Moon drops a hammer and a feather from the same height in vacuum. What happens?

(A) Hammer falls first because it has more mass
(B) Feather falls first due to less inertia
(C) They land together, neglecting initial differences
(D) Neither falls because the Moon has no gravity

Q.58 If local noon occurs at longitude 75° E, the approximate local solar time at longitude 90° E at that instant is:

(A) 10:00 am (B) 11:00 am
(C) 1:00 pm (D) 2:00 pm

Q.59 Why do observers on Earth usually see nearly the same face of the Moon?

(A) The Moon never rotates on its axis
(B) The Moon rotates once per orbit around Earth
(C) Earth's atmosphere hides the far side
(D) The Moon is stationary relative to the Sun

Q.60 Which contribution is correctly associated with ancient Indian astronomy?

(A) Aryabhata argued that Earth rotates about its axis
(B) Aryabhata discovered Neptune with a radio telescope
(C) Varahamihira launched India's first satellite
(D) Bhaskara II established the existence of black holes experimentally

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ANSWER KEY

CLASS IX | ETHEX 2027 SAMPLE QUESTION PAPER

PHYSICS

1. B	**2. B**	**3. A**	**4. D**	**5. A**	**6. B**	**7. B**	**8. C**	**9. C**	**10. C**

CHEMISTRY

11. A	**12. B**	**13. A**	**14. A**	**15. C**	**16. B**	**17. C**	**18. A**	**19. C**	**20. A**

BIOLOGY

21. B	**22. C**	**23. B**	**24. B**	**25. C**	**26. B**	**27. B**	**28. A**	**29. B**	**30. B**

MATHEMATICS

31. D	**32. C**	**33. B**	**34. C**	**35. B**	**36. B**	**37. D**	**38. B**	**39. A**	**40. B***

IQ / MENTAL ABILITY

41. A	**42. C**	**43. A**	**44. B**	**45. B**	**46. A**	**47. C**	**48. A**	**49. C**	**50. B**

ASTRONOMY

51. C	**52. B**	**53. B**	**54. B**	**55. B**	**56. C**	**57. C**	**58. C**	**59. B**	**60. A**

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